

9 Gas exchange

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Ventilation refreshes alveolar air and maintains oxygen and carbon dioxide partial-pressure gradients. Numerous alveoli provide a large surface, a thin epithelium short diffusion path and capillary networks carry gases away. Insects have tracheae and tracheoles delivering air toward cells, while stomata regulate plant gas exchange.

Worked example: follow the reasoning

If oxygen concentration differs by 10 units across an alveolar membrane versus 5, diffusion rate increases under otherwise identical Fick conditions.

Quick check · multiple choice

A What adaptation lowers diffusion distance at alveoli?

[1]

- A Thin epithelium
- B Thick cartilage
- C Small total area
- D No blood supply

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why pulmonary fibrosis reduces oxygen transfer even if alveolar oxygen partial pressure stays constant.

[3]

Answers and explanation

Quick check: A. Thin epithelium

Question 1 [3 marks]

Fibrosis thickens diffusion barrier [1]; diffusion distance increases [1], slowing oxygen uptake into blood [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Biology 9700, 2025–2027. <https://www.cambridgeinternational.org/Images/664560-2025-2027-syllabus.pdf>

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